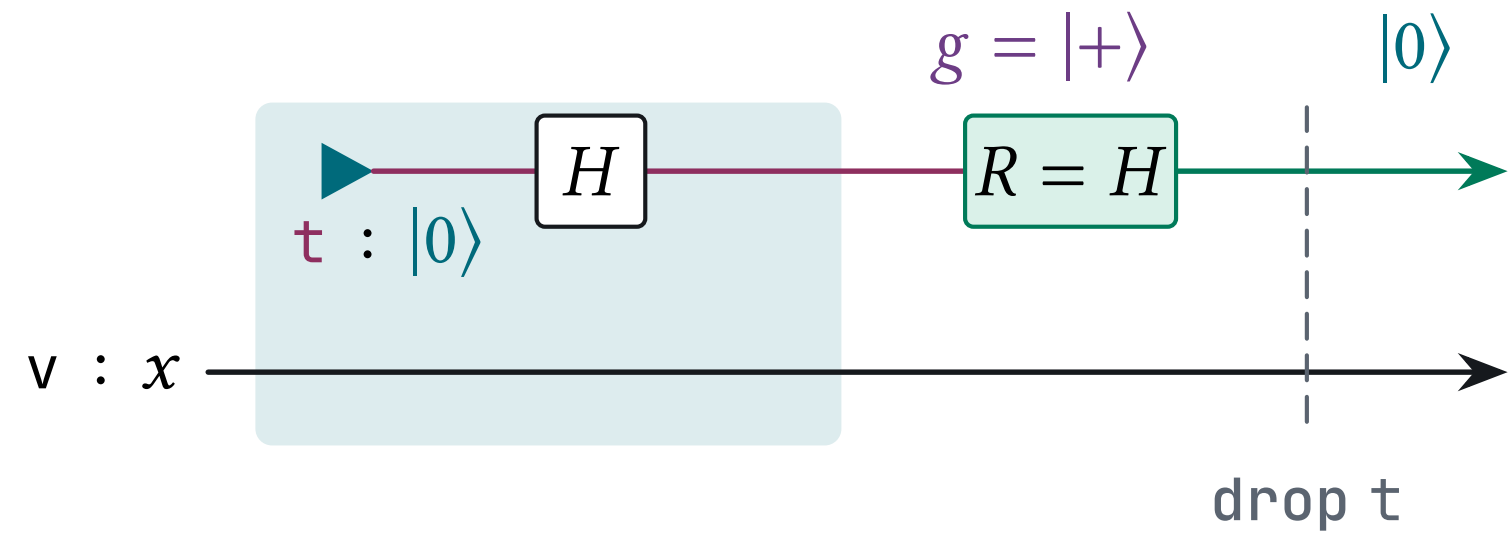$$K^{(a)}x = |+\rangle \otimes x$$